

Promptior Technical Test

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Links

Github repository:

<https://github.com/SantiagoBolumburu/langserve-chatbot>

Chatbot:

<http://langse-publi-zypqg3ybyucf-1111488589.us-east-1.elb.amazonaws.com/docs>

Project Overview

Approach:

I started by reading the suggested Langchain documentation, and the documentation about implementation of chatbots and RAG Chains/Agents linked on it, as well as other related documentation as it was being linked. Then I implemented said chatbot, the RAG chain and agent, by following along the documentation.

After that I decided on the implementation to follow would be an RAG Agent.

I implemented it similarly to the one on the documentation, but changing the data sources to the services section of the Promptor web site, as well as a text file with data I manually scraped from Promptor LinkedIn about section (which was the only source for the company's founding I could find).

Finally, I followed along the Langserve documentation to generate the project in which I placed the agent and finally deployed it to AWS.

Technologies

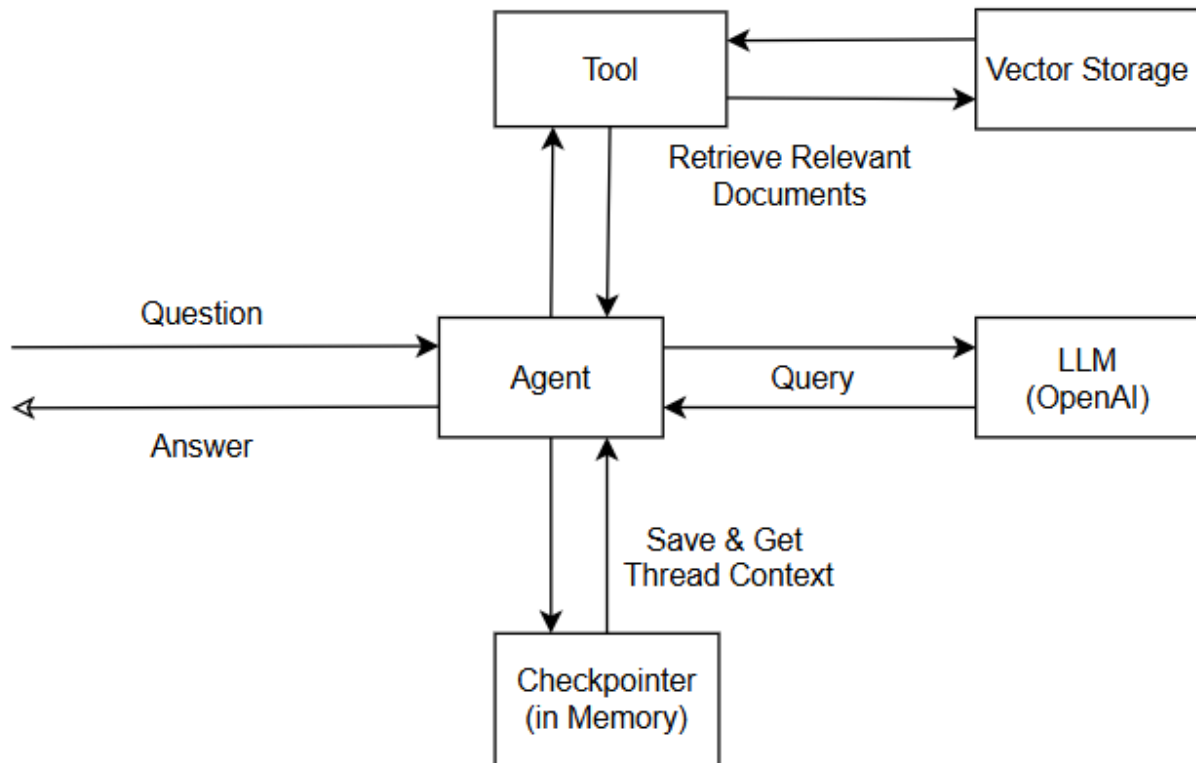
Technology	Purpose
OpenAI API	LLM model
Langchain	Development of the RAG Agent
Langserve + FastAPI	Development of the service that provides access to the RAG Agent
AWS	Cloud services to deploy the service

Challenges:

- Answering the question “When was the company founded?” proved a bit tricky since the only place where that information was available was on Promtior LinkedIn page, and that information couldn’t be retrieved by the “WebBaseLoader”, so the solution I found was to copy the information into a txt file and loaded it into a document from there.
- Langserve deployment required having Docker engine running on my local PC, although I already had Docker installed, this time for reasons I couldn’t figure out it didn’t start. So, the solution was to change the development environment from Windows to Linux (by using WSL (windows subsystem for Linux)) and download Docker there.
- Another problem was the deployment of AWS, which failed several times. Originally the project ran just fine on my PC, but that was because I was running it directly, after I built and ran the container locally, I found out that the problem was missing env variables and packages that weren’t adequately added.

Component Diagram:

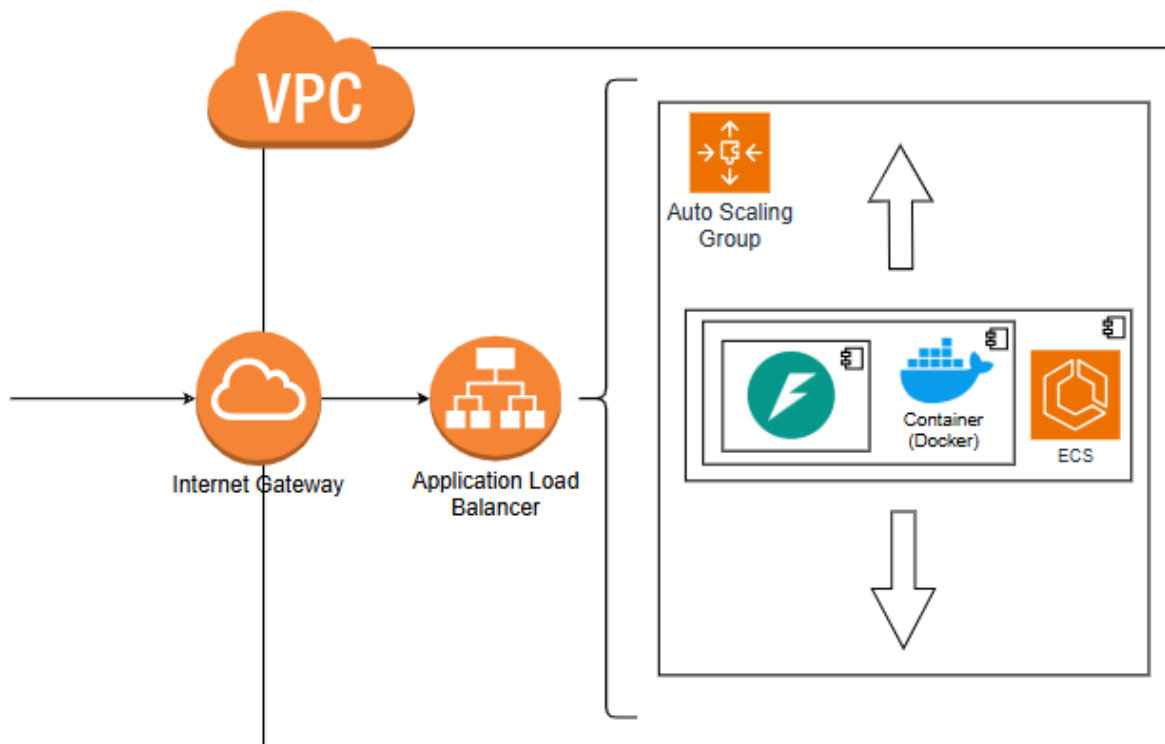
Agent:



The Agent receives a question, from then it can use tool (in this case the only tool is for retrieving documents) to get relevant information or get the thread context and make a query to the LLM, these 2 steps can be repeated multiple times, and at last it generates an answer.

If it was a Chain, one could state which steps it takes and in what order, but agents run the process several times as they see fit.

Infrastructure:



It has more components, this are the most relevant ones.